

Q Define Cost-benefit analysis (CBA). Explain the decision criteria of CBA.

Ans:

CBA ~~are~~ measures the benefits and costs of projects in money terms. This often requires that we place dollar values on years of life or improvements in health and well-being. These challenges have led to the development of new ideas, and health analysts now use the general term economic evaluation to represent the entire collection of tools.

Cost-benefit analysis addresses a wide range of measurement problems, often over many periods. It evaluates public investment costs, including those that have no markets to guide them. Likewise it investigates benefits that have no markets, to determine their prices. CBA also addresses public choices

involving either benefits or costs that are external to the market they came from.

CBA rests on the premise that a project or policy will improve social welfare if the benefits associated with it exceed the costs. These benefits and costs must include not only those directly attributed to the project but also any indirect benefits or costs through externalities or other third-party effects. Most simply, where B represents all the benefits and C represents all of the costs, a project is deemed worth-while if $B - C > 0$. We can also rank projects according to the benefit to cost (B/C) ratio; thus, a higher B/C ratio generally indicates a project that will deliver greater

social benefits, for a given dollar of costs.

~~decision~~ criteria of ~~costs~~

Q Why do we discount future value to into the present value? Explain.

Ans:

Multi-period projects require that we discount the future costs and benefits to put them on an equal basis with present values. This necessity arises from one of two basic reasons which suggest that future dollars usually are not worth as much to people as present-day dollars.

First, a dollar today could be used for something else. That dollar could have purchased an interest-bearing instrument such as bond, which would have been worth the dollar plus the interest in next period's dollars. Conversely, next period's dollar must be discounted at the market rate of interest to

discover its present value.

Second, many people tend to prefer the present when allocating spending. One consequence of this time preference is that the equilibrium interest rate will be positive, the reward for postponing consumption, and a cost for those who cannot wait or simply do not wish to.

Together, the time preference and the potential foregone interest from dollars spent on a project help explain why we discount future money values.

Q. How do we value life? Explain.

Ans:

Placing a value on human life is one of the most difficult but often unavoidable tasks in health care CBA. The first of several approaches, known as the human capital approach, estimates the present value of an individual's future earnings. This approach has been especially favored in legal applications that require estimates of damages. It also measures the loss of national output from mortality and morbidity or the production gains from saving and extending life.

willingness to pay and willingness to accept
Consider the question, "How much are you willing to pay for the reduction

in risks provided by new locks on your door?" The flip side of this concept is the compensation you would require to accept an additional risk to life and limb. The willingness to accept method also has many everyday examples. Its theoretical basis comes from the literature regarding the compensating differentials paid to laborers ~~across~~ across various lines of work.

Contingent Valuation:

It is often hard to derive a market test for willingness to pay for a risk-reducing medical treatment. Instead, the method of contingent valuation poses sets of medical contingencies such as: "If you faced an X% (high) risk of heart attack, how

much. Would you be willing to pay for a medical procedure that would reduce your ~~risk~~ risk to 1%?

The set of questions, visual images (if used), and researcher interactions is called the format. Contingent valuation has enabled practical studies to move forward, but it has proved vulnerable to challenge. Some researchers view problems of "hypothetical bias", the idea that the subject will not or cannot answer hypothetical answers questions realistically; changes in format can induce change in the valuation, a so-called framing bias.

How valuable is the last year of life?

The elderly spend over one-quarter of their total health-care expenditures

on the last year of life. Even economists using standard utility analysis question why one's utility of life would get to be so high when one has low natural prospects for living many years and when one's motive to bequeath to one's heirs is a ready alternative.

Becker, Murphy and Philipson offer an interesting resolution to this puzzle. They begin by arguing against linearity in the calculation of the value of a life-years.

Their analysis rests on four ~~int~~ intuitive ideas. First, for many of the very old and sick, their resources have very low opportunity costs because they cannot enjoy their wealth once they have died. Second,

they may rationally have "hope" for living, including the hope that more advanced health care will be developed within their extended life time. Third, their social value of life may be very high. Finally, these authors show that the value of an extended life-year may be as high for frail patients as it is for those with better health.

Q Define cost-benefit analysis. Explain the basic principles of CBA.

Ans:

CBA : ଆବଶ୍ୟକ ପ୍ରଶ୍ନ

Q

Basic principles of CBA:

The basic idea of cost-benefit analysis mirrors the measurement problems that it addresses. For example, CBA evaluates public investment costs, including those that have no market. Likewise, it investigates benefits that have no markets, to determine their prices. These may entail benefits or costs over many periods. CBA also addresses public choices involving either benefits or costs that are external to the market they come from. Projects,

such as putting a dam across a river or reconsidering the efficiency of a congressional program, are often controversial. Within health economics, controversies over the wisdom of immunization programs, patient screening, or heart transplants, for example, involve many problems and criticisms that cost-benefit thinking must address. The benefits and costs that are counted must include not only those directly attributed to the project but also any indirect benefits or costs through externalities or other third-party effects. Thus, where B represents all the benefits and C represents all of these costs, a project is deemed worthwhile if $B - C > 0$. We can also rank projects

according to the benefit to cost
(B/C) ratio; thus, a higher B/C
ratio generally indicates a project
that will deliver greater social benefits
for a given dollar of costs.

Q What are the limitations of CBA?
Explain.

Ans:

Some limitations of Cost-Benefit analysis:

1. There are a lot of benefits that are hard to quantify. And a lot of cost items that are considered as un-measurable. So, the chances of inaccuracies are great, due to the lack of true estimated value.
2. In case of analysis the main executive may relies on previous data and information, but the data and information are ever-changing in the current scenarios.
3. Cost-benefit analysis works best in perfectly competitive market with a

large numbers of service holders and service providers. But besides this, there are monopolies and the government whose interventions distort the market policy.

4. Sometimes the cost-benefit analysis are calculated for a specific period of time, and it causes serious miscalculation in present value. This leads to inaccurate analysis.

5. For very large projects with a long-term time horizon, a cost-benefit analysis might fail to account for important financial concerns such as inflation, interest rates, varying cash flows, and the present value of money.

6. A cost-benefit-analysis requires that all benefits and costs be identified and appropriately quantified. Unfortunately, human error often results in common cost-benefit analysis errors such as accidentally omitting certain costs and benefits and ambiguity monetary price of services, which can lead to increased risk and inefficient decision-making.

Q Define Cost-Benefit - Analysis .

Ans: Previous question

Q What is Discounting? Why do we discount future value into its present value? Explain.

Ans: Previous question

Ex) Consider a health care project that costs \$10,000 today. It will provide benefits of \$4,000 at the end of year 1, \$3,500 at the end of year 2, and \$3,500 at the end of year 3. If the discount rate is 6%, will this health care project be approved using CBA? Would your answer change if the discount rate is

i) 5% and

ii) 4%